

Kinematic Design and Motion Simulation of a Four-Legged Jansen Walking Mechanism

Planar Eight-Bar Linkage Assembly, SolidWorks Motion Simulation, and Leg-Trajectory Characterization

1. Project Overview

This project covers the assembly and motion simulation of a four-legged walking base built on the Jansen linkage — a planar eight-bar leg mechanism originally developed by Theo Jansen for his kinetic “Strandbeest” sculptures. The defining property of the linkage is that a single rotating crank drives a chain of ternary and binary coupler links whose lowest point traces a closed, foot-shaped path: a long, nearly flat segment that acts as the ground-contact “stance” phase, followed by a fast, high-clearance “swing” arc that returns the foot forward. No ankle, hip actuator, or electronic gait control is required — the entire walking motion is encoded directly in the linkage geometry.

The model built for this project replicates that single-leg linkage four times around a shared frame and crank to form a complete four-legged base, matching the construction sequence set out in the accompanying course tutorial. The assembly was driven in SolidWorks Motion by a rotary motor on the crank, and the resulting foot trajectory was captured quantitatively — not just animated — by extracting linear displacement and velocity time histories for the lower ternary link (the foot reference point) on two of the four legs.

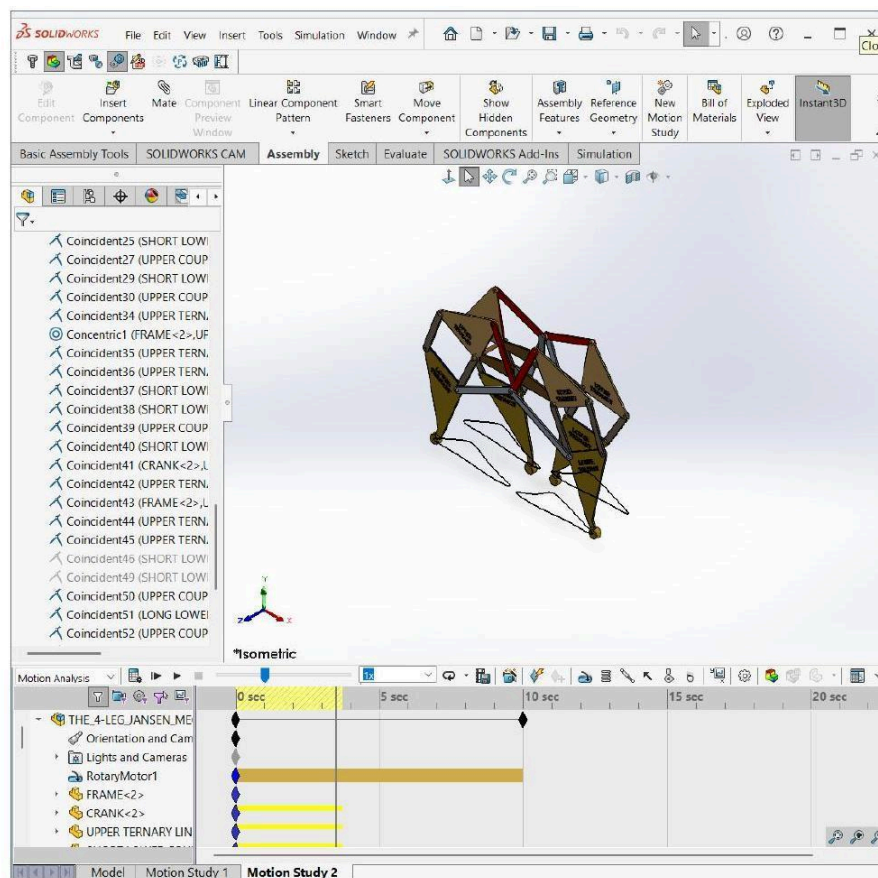


Figure 1. Four-leg Jansen assembly in SolidWorks, shown mid-simulation with the coincident/concentric mate list and Motion Study (RotaryMotor1) timeline.

2. Design Objectives

- Assemble the seven unique Jansen linkage components (frame, crank, two ternary links, three coupler links) using coincident and concentric mates, replicated across all four legs.
- Drive the assembly with a single rotary motor at the crank, matching the tutorial's 60 RPM (1 rev/sec) input speed.
- Generate trace curves on the foot point of each lower ternary link to visualize the characteristic Jansen gait curve for all four legs simultaneously.
- Extract linear displacement (X, Y, Z) and linear velocity (X, Y, Z) time-history plots for representative legs to quantitatively verify the gait shape, leg-to-leg phasing, and planarity of the mechanism.
- Interpret the resulting kinematic data in terms of the mechanism's walking function — stance/swing timing, leg pairing, and confirmation that the linkage remains a single-degree-of-freedom planar system.

3. Mechanism Architecture

Each leg of the mechanism is a classic Jansen eight-bar linkage: the frame (ground) and crank (input) are shared across the whole base, while five additional link types — the upper ternary link, lower ternary link, long lower coupler, short lower coupler, and upper coupler — are repeated once per leg. With four legs built from these five link types, the walking base comprises on the order of twenty leg-forming link instances in addition to the shared frame and crank, all fully constrained by coincident and concentric mates so that the entire four-legged assembly still moves with a single degree of freedom driven from one crank.

4. Motion Study Setup

The simulation was configured in the SolidWorks Motion add-in. A rotary motor was placed on the front face of the crank, oriented counter-clockwise, and driven at a constant 60 RPM (one full revolution per second) — the same reference speed used throughout the course tutorial. The Motion Study timeline (visible in Figure 1, alongside the full coincident/concentric mate list constraining the linkage) was run out to capture several full gait cycles.

Trace curves were then generated on the circular foot edge of each lower ternary link using the Results and Plots → Trace tool, producing the closed “foot curve” for all four legs at once. From there, Linear Displacement and Linear Velocity plots were extracted for the same foot reference points so the qualitative trace shape could be backed by quantitative time-history data.

5. Results — Displacement Characterization

Figure 2 shows the linear displacement time histories for two representative legs. The X-component plots (Plot7 / Plot11) are clean, near-sinusoidal curves that mirror one another — one leg's X-displacement runs from -7 to -73 in while the other runs from 7 to 73 in over the same period — confirming that the two legs are driven exactly out of phase, as required for a stable four-legged gait rather than all legs stepping in unison.

The Y-component plot (Plot13) is where the Jansen linkage's design intent becomes visible: rather than a smooth sinusoid, the curve holds a long, nearly flat plateau between -80 and -76 in — the stance phase, where the foot stays close to constant height while in contact with the ground — before spiking sharply up to -62 in and back down, the fast swing phase that lifts and advances the foot. This asymmetric shape is exactly the behavior the linkage was originally optimized to produce.

The Z-component plot (Plot14) stays essentially flat at a constant value with only minor numerical ripple. Because Z is the direction perpendicular to the plane of the leg linkage, this result confirms the mechanism is behaving as intended: a fully planar, single-degree-of-freedom linkage with no lateral drift or twisting introduced by the assembly mates.

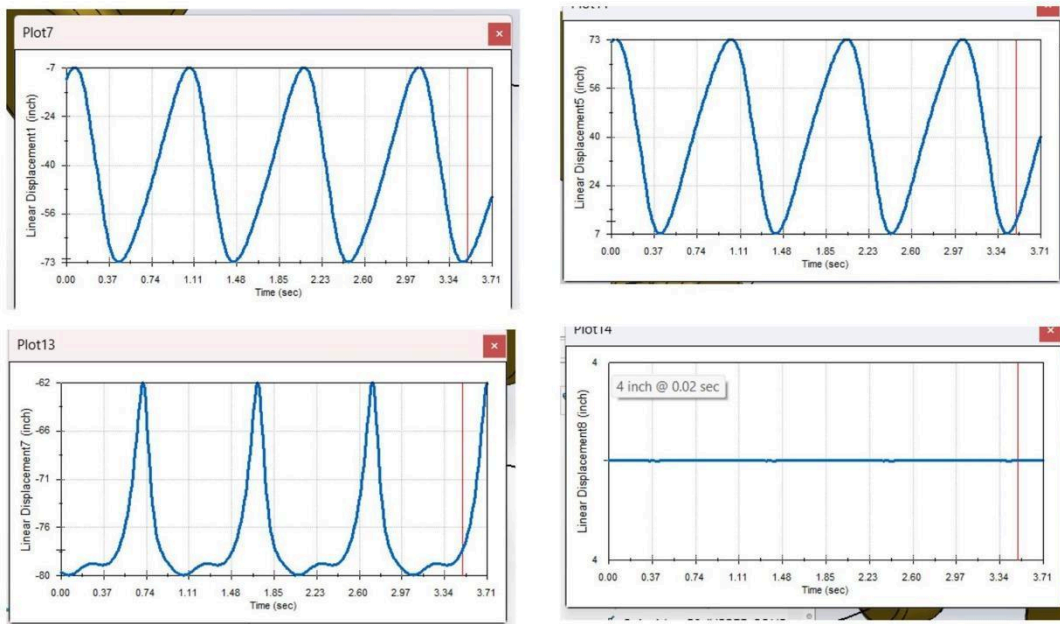


Figure 2. Linear displacement time histories: X-component for two legs (Plot7, Plot11), Y-component stance/swing curve (Plot13), and Z-component confirming planar motion (Plot14).

6. Results — Velocity Characterization

Figure 3 shows the corresponding velocity time histories. The X-velocity plots (Plot15 / Plot16) reach a shared peak of about 115 in/sec forward and a much larger 276 in/sec during the return stroke — the same out-of-phase pairing seen in the displacement data is preserved in velocity, with the two curves offset by roughly half a cycle.

The Y-velocity plot (Plot17) is the clearest confirmation of the stance/swing behavior: velocity stays low and gently oscillating during the long stance plateau, then drops sharply to a peak of -198 in/sec exactly where the Y-displacement plot showed its steep swing-phase transition. This is the expected result of differentiating a curve that is mostly flat with one fast transition per cycle.

The Z-velocity plot (Plot18) remains at essentially zero throughout the simulation, reinforcing the Z-displacement result from Section 5: the leg motion has no meaningful out-of-plane velocity component, which is the expected outcome for a correctly constrained planar linkage.

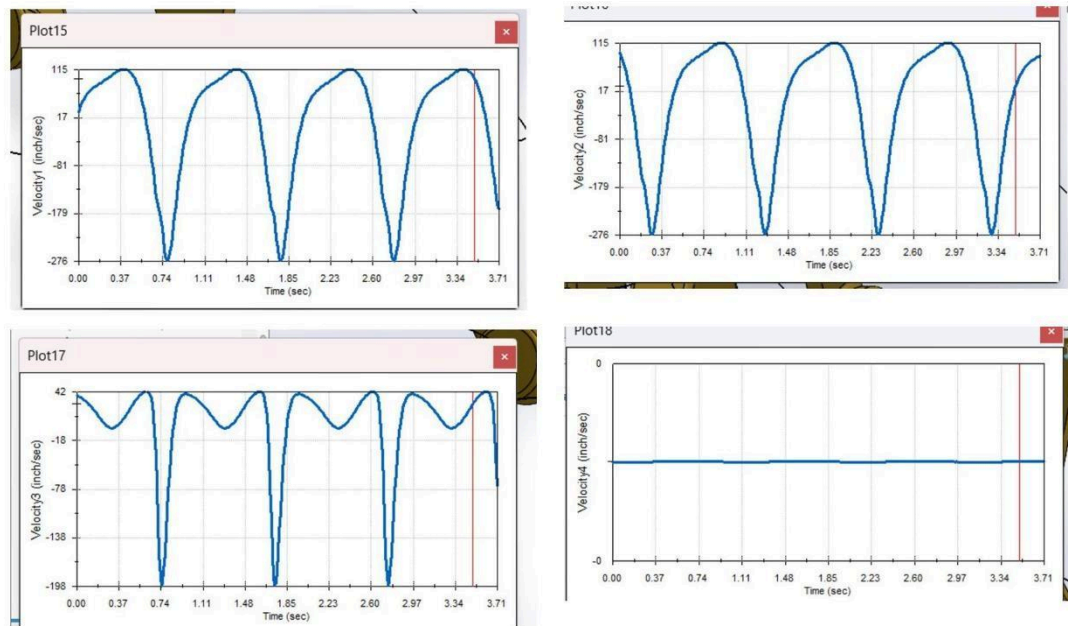


Figure 3. Linear velocity time histories: X-component for two legs (Plot15, Plot16), Y-component swing-transition spike (Plot17), and Z-component confirming near-zero out-of-plane velocity (Plot18).

Across every plotted quantity, the cycle period measures approximately 1.1 seconds — consistent with the 60 RPM (1.0 second per revolution) motor input, with the small difference attributable to the sampling resolution of the exported plots rather than an error in the model or motor setup. Table 2 summarizes the full set of extracted results.

7. Engineering Discussion

The results validate the mechanism on two levels. First, kinematically: the flat-stance / fast-swing Y-displacement shape, the near-zero Z-displacement and Z-velocity, and the clean sinusoidal X-motion together demonstrate that the assembly reproduces the Jansen linkage's signature planar foot curve rather than an arbitrary or under-constrained motion — any meaningful drift in the Z-plots would have signaled a mis-mated or over-constrained joint somewhere in the twenty-link assembly.

Second, at the system level: the opposite-phase X-displacement and X-velocity pairing between the two examined legs confirms the four-leg base is timed for a genuine walking gait rather than four legs stepping identically. Because every leg shares the same crank and frame, this phase relationship is set purely by how each leg's linkage is oriented and mounted around the shared drive — a design decision that this quantitative motion analysis was able to confirm numerically rather than only by eye from the animation.

8. Deliverables Captured

- Shaded isometric view of the full four-leg Jansen assembly with active trace curves and the Motion Study / mate tree visible (Figure 1).
- X-component linear displacement plots for two representative legs, confirming out-of-phase pairing (Figure 2, Plot7 / Plot11).
- Y-component linear displacement plot showing the characteristic stance/swing foot curve (Figure 2, Plot13).
- Z-component linear displacement plot confirming planar motion (Figure 2, Plot14).

- X-component linear velocity plots for the same two legs, confirming the phase relationship carries through to velocity (Figure 3, Plot15 / Plot16).
- Y-component and Z-component linear velocity plots corroborating the stance/swing transition and planarity results (Figure 3, Plot17 / Plot18).

9. Conclusion

This project took the Jansen leg linkage from a set of individual SolidWorks parts through a fully mated, motor-driven, four-legged walking assembly, and then backed up the visual motion study with quantitative displacement and velocity data. The extracted plots confirm the three properties that make the Jansen linkage useful as a walking mechanism in the first place: a smooth, low-vertical-bob stance phase, a fast forward swing phase, and strictly planar single-degree-of-freedom motion — all driven from one rotating crank with no additional actuators. The same workflow — mate-driven assembly, motor-driven motion study, and results extraction — mirrors how a mechanism designer would validate a legged or walking linkage concept before committing it to a physical prototype.